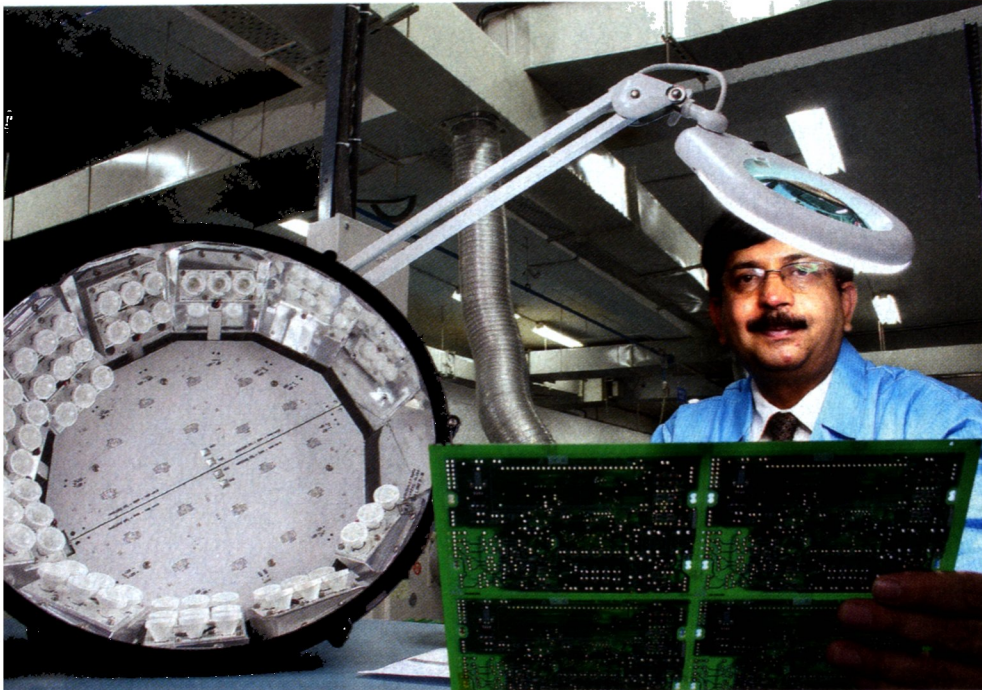


SANJAY RAMACHANDRAN



GREENOLITE

Entrepreneur: Sanjay Bhardwaj
Investors: Israeli and American private equity players
IDEA: Solar-based home, street lighting systems

to the company, is the highest energy producing solar panel in the market today. What is unique about this start-up is that they make lead-free modules. At the core of HHV team's work lies a technology which involves depositing silicon on glass to turn it into an electricity-generating module which creates the high efficiency.

Unlike in the past, start-ups in clean technology are not coming in with just the facilities to generate clean energy and reap the benefits being doled out by the Government. Like Ameet Shah and Saurabh Sen who invested their own capital to start Astonfield Renewable Resources in 2005 and

are bringing with them world class technology. Astonfield's bulk investments are in solar with four solar pv plants in West Bengal, Rajasthan, Gujarat and Karnataka. Astonfield's tie-ups with Beck Energy of Germany and Isolux Curson of Spain for their solar projects allow Shah and Sen to depart from the traditional solar pv players and deploy 2 MW inverters which can effectively curb volatility in the grid. India's grids suffer from high volatility because of the low quality of power.

With solar becoming a mantra for cleantech companies, start-ups are also diversifying into solar-based LED lighting. Akshat Khare who entered this segment with Pegasus Semiconductor, keeping in mind the growing demand for alternatives, feels his gamble has paid off. Pegasus makes off-grid home and street lighting systems using LED lights and solar power. It

has done about 1,200 installations in Barmer, Rajasthan, and about 35 with companies such as Zydus Cadilla and other government bodies and is expecting to reach a revenue of Rs 3 crore by the fiscal-end. Sanjay Bhardwaj's GreenOlight, a Bangalore-based company, leverages the solid-state technology of LEDs to provide lighting solutions, exploring lighting fixtures in new constructions and retail avenues in LED retrofits.

In a country where recycling e-waste is managed in the unorganised sector, there is a huge opportunity in organising the activity and that is what Nitin and Rohan Gupta discovered. Attero Recycling, which raised \$6.3 million from venture capital firms NEA-IndoUS Venture and Draper Fisher Jurvetson, used the money to develop a state-of-the-art integrated e-waste recycling facility at Roorkee which has the capacity to recycle 36,000 tonnes a year. While most unorganised recyclers manage to extract only a few of the materials, Attero's plant is equipped to extract 98 per cent of materials such as copper, lead, nickel, zinc, plastic, iron, silver and even gold to an extent. To collect e-waste from various sources, Attero has signed up third party logistics provider Vikram Logistics, which will pick up even one personal computer from homes, as well as scrap from other sources. The company has collected 300-400 tonnes of e-waste and plans to do about 1,000 tonnes a month initially and reach 3,000 tonnes a month in the next four-five years.

It is obvious that the tide has turned irrevocably and positioned India to take on the cleantech investment in a big way. Once the companies are convinced that the investor-friendly regime is here to stay, it will pave the way for entry of their manufacturing units in India, and that advantage translates into better technologies for India, high-end manufacturing base and most importantly jobs. ■

Organised e-waste recycling is the panacea for the growing problem of hazardous dumping.

NITIN GUPTA, Co-founder, Attero Recycling

